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Full length article

Exploring embodied greenhouse gas (GHG) emissions of mass timber construction: A comparative study of life cycle assessment databases

Zhuocheng Duan *, Hossein Omrany, Jian Zuo

School of Architecture and Civil Engineering, The University of Adelaide, Australia

A R T I C L E I N F O

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A B S T R A C T

With global efforts in reducing building operational energy consumption, attention is increasingly turning toward mitigating embodied greenhouse gas (GHG) emissions from building materials manufacturing and processing, particularly for emerging materials like mass timber. Quantifying embodied emissions typically requires the use of GHG emission coefficients from life cycle assessment (LCA) databases. However, significant variability persists in the results of LCA studies while their underlying causes remain largely unexplored. Therefore, this research aims to compare embodied GHG emissions data of mass timber across five widely used LCA databases (Ecoinvent, AusLCI, EPiC, ICE, and ÖKOBAUDAT). Through detailed examination at material and building levels, the research reveals significant methodological inconsistencies that create substantial variations in embodied emissions. At the material level, mass timber emissions span from -686.80 to 1718.00 $kgCO_2eq/m^3$. Building-level analysis (A1-A3) reveals emission ranges of -170.31 to 434.62 $kgCO_2eq/m^2$ for mass timber construction and 69.79 to 485.95 $kgCO_2eq/m^2$ for concrete buildings. The study identifies several challenges in current mass timber LCAs, including inconsistent treatment of biogenic carbon flow, varying system boundary definitions, and inadequate consideration of End-of-Life (EoL) processes for mass timber. Regional variations in manufacturing processes, energy sources, and transportation distances emerge as essential factors influencing data quality. This highlights the need for establishing standardized approaches for biogenic carbon accounting, implementing detailed material documentation systems, and developing more comprehensive and transparent databases to enhance the reliability of mass timber LCA.

1. Introduction

The building sector is responsible for over one-third of the total energy consumption and associated carbon emissions worldwide (UNEP, 2024). Despite growing attention to sustainability in construction, emissions attributed to the building sector still increased by 1 % in 2022 (IEA, 2024), reaching new highs of about 10 $GtCO_2$ (UNEP, 2024). These substantial emissions contribute significantly to global warming, resulting in increased average temperatures and more frequent extreme weather events that threaten human wellbeing and the built environment.

Building environmental impacts involve the emissions from building operations, as well as the embodied emissions arising from the production and processing of materials (Röck et al., 2020). As buildings worldwide become more energy-efficient and global efforts progress toward a carbon-neutral building sector, focus has increasingly shifted to the embodied emissions of building materials (Röck et al., 2023). These promote the exploration of alternative construction materials and techniques that can reduce the embodied impact of buildings (Pomponi and Moncaster, 2016).

The use of mass timber has become an emerging area in building practices to limit greenhouse gas (GHG) emissions due to its lower carbon emissions compared to carbon-intensive materials as well as carbon sequestration (Younis and Dodoo, 2022). The mass timber

Abbreviations: GHG, Greenhouse Gas; CLT, Cross-Laminated Timber; LCA, Life Cycle Assessment; RC, Reinforced Concrete; EoL, End-of-Life; GLT, Glued Laminated Timber; LVL, Laminated Veneer Lumber; LCIA, Life Cycle Impact Assessment; GWP, Global Warming Potential; EPD, Environmental Product Declaration; GGBFS, Ground Granulated Blast Furnace Slag; EPS, Expanded Polystyrene; XPS, Extruded Polystyrene; HFC, Hydrofluorocarbon; MUF, Melamine-Urea-Formaldehyde; PUR, Polyurethane; LOSP, Light Organic Solvent Preservatives; CCA, Copperchrome-arsenate; BIM, Building Information Modeling; LOD, Level of Development.

* Corresponding author.
E-mail address: zhuocheng.duan@adelaide.edu.au (Z. Duan).

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includes engineered wood products like cross-laminated timber (CLT), glued laminated timber (GLT or Glulam), and laminated veneer lumber (LVL) (Ayanleye et al., 2022). These materials can be viable alternatives to conventional structural materials such as concrete and steel due to their high strength-to-weight, faster building construction time, fire resistance, and environmental benefits (Ramage et al., 2017). Among these, CLT is the most widely engineered wood product that consists of uneven layers of glued boards or planks, with each layer placed at 90 degrees to each other (Swedish Wood, 2019). The advent of mass timber has enabled the construction of high-rise wooden buildings previously thought impossible. For instance, the MJØSTÅRNET building completed in March 2019 in Norway reached 18-storey (85.4 m) with glulam and CLT structures (Liven and Abrahamsen, 2023). These achievements have prompted countries to revise their building codes to allow taller timber construction, such as in Europe (Dodoo et al., 2014a), North America (Pierobon et al., 2019), Australia (Jayalath et al., 2020), and China. The adoption of mass timber enables buildings and cities to function as temporary carbon sinks. By sequestering carbon during their service life (typically 50–100 years), mass timber construction may help provide time for the development and implementation of permanent carbon capture and sequestration technologies in the future (Churkina et al., 2020).

Understanding the environmental impacts of mass timber construction is particularly important as the benefits of carbon storage and reduction must be weighed against emissions from manufacturing, transportation, disposal and other life cycle processes. Life cycle assessment (LCA) provides a standardized method to evaluate these environmental impacts throughout the entire building life cycle (Cabeza et al., 2014; Chau et al., 2015). LCA framework includes different phases such as life cycle inventory and impact assessment. To quantify the inputs and outputs, data collection is needed for building materials. Databases are developed to quantify the embodied impacts of products and materials, with databases exhibiting significant variation in their geographical scope, sectoral focus, and methodological approaches.

Previous studies have discussed the inconsistencies of embodied emissions and explored the reliability and consistency challenges in LCA databases. For instance, Martínez-Rocamora et al. (2016) evaluated ten databases based on their methodology, data quality and comprehensiveness. In another study, Teng et al. (2023) analyzed the flexibility, comprehensiveness, transparency, and uniformity of 20 databases. While these analyses identified significant variations in embodied carbon data, they primarily focused on traditional construction materials. Bio-based materials, such as mass timber, pose unique challenges for LCA calculations due to their distinct characteristics, particularly concerning biogenic carbon flow and End-of-Life (EoL) scenarios (Arehart et al., 2021; Mouton et al., 2023; Pawelzik et al., 2013; Peñaloza et al., 2016). While data inconsistencies across LCA databases are identified for conventional building materials, their implications for mass timber assessment remain understudied. The reliability and consistency of embodied emissions depend heavily on the quality and comparability of the underlying data (Anand and Amor, 2017), particularly for mass timber construction assessments.

Therefore, this research aims to analyze how different LCA databases address certain complexities of mass timber, including carbon storage accounting, allocation methods, and system boundary definitions. However, it is noteworthy to mention that this study does not aim to compare the magnitude of emissions across different databases. Instead, it seeks to explore the underlying methodological differences, technical assumptions and calculation challenges that drive these inconsistencies. By doing so, this study will provide insights into selecting datasets and assessing the embodied impacts of mass timber.

2. Literature review

In recent years, there has been a surge in research on the LCA of mass timber construction, driven largely by the goal of achieving carbon neutrality in the building sector (Younis and Dodoo, 2022). One popular area lies in exploring the potential environmental benefits of mass timber construction through comparative analyses with conventional buildings (Table A1). These studies reveal varying degrees of carbon reduction across different applications and contexts. For instance, Kang et al. (2024) showed that substituting concrete walls with CLT elements achieved a 14.42 % reduction in carbon emissions. Liu et al. (2016) also reported that CLT buildings offered a 40 % reduction to concrete buildings from cradle to grave in the cold region of China. In another study, Pierobon et al. (2019) investigated embodied impacts of two fireproofing and charring designs of the hybrid CLT buildings with concrete buildings, with an average 26.5 % decrease in global warming potential (GWP). The multi-criteria decision-making framework was used to evaluate CLT, reinforced concrete (RC), and steel structures across different indicators (Invidiata et al., 2018). Their analysis revealed that RC frames performed better in thermal comfort and cost metrics, whereas concrete and steel frames showed advantages in energy demand. They also reported that CLT buildings achieved lower carbon emissions compared to other alternatives. Hart et al. (2021) conducted a comprehensive analysis of whole-life embodied carbon in multistory buildings, comparing steel, concrete, and timber structures. Their findings revealed that while upfront emissions dominate the embodied carbon, post-construction emissions remain significant, particularly for timber structures, where an average of 36 % of total emissions occur after the construction phase.

The research scope has expanded beyond simple material substitution to encompass comparative analyses of various timber structural systems. For instance, Balasbaneh and Sher (2021) conducted a comparative analysis of different mass timber buildings, examining CLT and GLT manufactured from hardwood and softwood. Their findings indicated that CLT exhibited lower environmental impacts in terms of GWP, ozone layer depletion (OLD), land use, and terrestrial ecotoxicity, whereas GLT showed lower impacts for human toxicity potential (HTP) and fossil depletion potential (FDP). Dodoo et al. compared three building systems: CLT, beam-and-columns, and modular buildings in terms of GHG emissions (Dodoo et al., 2014a) and primary energy (Dodoo et al., 2014b). Notably, they demonstrated that CLT low-energy building exhibited the lowest carbon footprint, whereas conventional beam-and-column building resulted in significantly higher carbon emissions (Dodoo et al., 2014a). Tettey et al. (2019) discussed the life cycle primary energy of two building standards of concrete, CLT and modular timber buildings in Sweden.

As buildings achieve greater operational energy efficiency, attention has increasingly shifted to embodied emissions from materials processing. This shift in focus reflects the growing recognition that embodied impacts can represent up to 50 % of whole life GHG emissions in energy-efficient buildings, and surpasses 90 % in extreme cases (Röck et al., 2023, 2020). According to a comprehensive study analyzing over 400 residential buildings (Zhang et al., 2023), approximately two-thirds of the total embodied carbon originates from structural materials. This significance is particularly evident in mass timber applications, where methodological challenges arise from complex biogenic carbon dynamics and system boundary definition. In the analysis of a hypothetical 43-story building in Australian context, Li et al. (2019) examined embodied emissions (A1-A3) across three scenarios: concrete construction (195 $kgCO_2eq/m^2$), mass timber non-structural elements (80 $kgCO_2eq/m^2$), and CLT with a concrete core (−311 $kgCO_2eq/m^2$). The great variation in results, particularly the negative emissions value, highlights the critical influence of methodological choices in assessment outcomes. Andersen et al. (2022) explored biogenic carbon dynamics in mass timber construction. Their base scenario, which excluded biogenic carbon dynamics, showed emissions of 903.7 versus 454.2 $kgCO_2eq/m^2$ for concrete and mass timber respectively. Including biogenic carbon resulted in values of 892.9 versus 288.5 $kgCO_2eq/m^2$, demonstrating how accounting choices significantly influence results. In another research, Peñaloza et al. (2016) developed comprehensive carbon flow models for CLT buildings, integrating three crucial aspects: biogenic carbon storage, forest management practices, and EoL scenarios. EoL processes, including reuse, recycling, incineration, or landfilling, not only influence biogenic carbon release but also create potential issues of double counting and assessment accuracy of the embodied impacts.

To support embodied impact assessment, various LCA databases have been developed with different geographical and sector focuses (Martínez-Rocamora et al., 2016). However, the diversity and variations among these databases have introduced new methodological challenges around selection and application of datasets. Previous studies have raised concerns about database quality assessment and practical implications of database selection. For instance, Teng et al. (2023) analyzed the flexibility, comprehensiveness, transparency, and uniformity of 20 databases, reporting that Ecoinvent, AusLCI, and Global Digital EPD performed better regarding the four-principle framework. Martínez-Rocamora et al. (2016) discussed ten databases from six perspectives, including scope, completeness, transparency and comprehensiveness, highlighting the Ecoinvent and Gabi as the most complete databases. The practical implications of database selection have also been demonstrated through empirical studies. Five databases (i.e., GaBi, Ecoinvent, IBO, CFP, Synergia) of the carbon emissions of the material stage for three reference buildings were compared (Takano et al., 2014b). The study reported high relative standard deviation for insulation and plastic materials in all cases. Mohebbi et al. (2021) reported that using the data from a detailed database led to a ~35 % decrease in embodied GHG emissions when compared to a more generic database.

In addition to the aforementioned challenges, the magnitude of emissions variations has also emerged as a significant concern. This is echoed in the findings of Pomponi and Moncaster (2018) that revealed substantial variations in embodied carbon values, with ranges of 284–1044 % across structural materials. While production processes and regional energy mixes contribute to these disparities, the two order of magnitude differences suggest fundamental methodological challenges. These challenges associated with LCA estimation have become particularly significant for mass timber construction, where multiple factors complicate environmental impact calculations. Specifically, variations arise from inconsistencies in LCA methods, GWP calculation approaches, biogenic carbon accounting, system boundary definitions, and uncertainties in recycling scenarios (Ouellet-Plamondon et al., 2023; Rasmussen et al., 2021; Younis and Dodoo, 2022). The significance of these methodological variations is highlighted by Moncaster et al. (2018), who identified that methodological disparities could result in greater, with a factor of 6.5 for CLT and 11.5 for steel buildings when comparing the lowest and highest embodied values. This is also consistent with the findings of Rasmussen et al. (2021) who reviewed 81 EPD of timber products, and reported significant variation in GWP, influenced by product density and consideration of EoL scenarios.

The implications of database selection extend beyond theoretical concerns to practical implementation challenges. Employing data from an inappropriate database/dataset may lead to inaccuracy in quantifying the embodied carbon, which can ultimately influence decision-making. Loli et al. (2023) demonstrated that product-specific EPD data yielded approximately 15 % lower GWP values compared to generic Ecoinvent data in a Norwegian case study (with the exception of solar panels). This finding is further supported by Kyaw et al. (2024), who demonstrated that substituting site-specific datasets for generic environmental impact databases (Ecoinvent) resulted in substantial reductions (up to 95 %) in GWP calculations. These studies collectively emphasize the critical importance of appropriate dataset selection and the potential for significant variations in LCA results depending on data source specificity. This is particularly problematic for mass timber construction, where the structural material choice substantially influences the overall embodied impacts. Databases commonly serve as primary data sources for mass timber LCA studies; however, systematic analysis of database selection implications on the embodied impacts of mass timber still remains largely unexplored.

While existing literature offers valuable insights into the carbon reduction potential of mass timber construction, a significant research gap remains in addressing methodological variations across LCA databases and their impact on assessment outcomes. Previous comparative studies (Lasvaux et al., 2015; Takano et al., 2014b; Teng et al., 2023) have discussed variations in LCA database results. However, these analyses have neither specifically addressed mass timber products nor thoroughly investigated factors and implications of such variations. This research gap becomes increasingly significant given three emerging factors: the rapid evolution of mass timber in databases, the unique complexities of biogenic carbon flow accounting, and the expanding geographical contexts of mass timber applications.

The objective of this research paper is to examine embodied GHG emissions data for mass timber construction through comparative analysis of five recognized databases: Ecoinvent, AusLCI, EPiC, ICE, and ÖKOBAUDAT. Key areas of investigation include: (1) diverse approaches to biogenic carbon accounting in timber products; (2) methodological overlaps leading to potential double-counting in emissions and sequestration calculations; (3) disparities in allocation methods of recycling module D that create varying interpretations of environmental impacts; and (4) fundamental differences in system boundary definitions, particularly associated with EoL scenarios. By analyzing these geographically diverse databases, this research not only quantifies the extent of variability in mass timber embodied GHG emissions data but also, more importantly, identifies the underlying factors contributing to these variations. These insights will inform efforts to enhance the reliability and comparability of mass timber construction LCA. The outcomes of this research will contribute to standardizing assessment methodologies and promoting the effective implementation of mass timber as a low-carbon building solution.

3. Materials and methods

3.1. Databases

The selected databases (Ecoinvent, AusLCI, EPiC, ICE, and ÖKOBAUDAT) provide mass timber datasets and have been widely used in previous studies (Table A1). As such, they were chosen to represent diverse geographical contexts and methodological approaches. Mass timber has been developed and widely adopted in Europe with Northern European countries such as Sweden and Norway leading in mass timber production and construction. The European context is represented by the Ecoinvent database, which is a comprehensive LCI database covering a wide range of construction materials and processes. Mass timber construction, particularly CLT applications, is gaining attention in Australia, with several notable projects completed in recent years. The Australian context is captured by the AusLCI and EPiC databases. Similarly, the ICE database provides building material datasets specific to the UK construction context, and ÖKOBAUDAT provides data tailored to the German construction sector. Table 1 provides detailed information regarding these databases. The datasets were collected by 20 November 2024. In this paper, carbon data is the indicator considered, as they represent the most widely used metric for impact assessment. While other databases (e.g., Ecoinvent, ÖKOBAUDAT) include multiple indicators (e.g., energy use, ozone depleting potential (ODP), acidification potential (AP)), AusLCI and ICE databases exclusively focus on carbon metrics.

3.2. Data collection and normalization

From each database, we extracted data for the common building materials. The analysis focused primarily on two mass timber products: CLT and Glulam (some called GLT), focusing on their universal application in building structures. The extracted data included GWP or embodied carbon values in $kgCO_2eq$ per functional unit, system

Table 1
Comparison of selected LCA databases.

Database	Version	Region	Sector	Date	Indicators	Mass timber	Biogenic carbon	EoL	System boundary	Land Use Change	Access
Ecoinvent	3.8	Global, but mainly focuses on Europe and Switzerland	Multi-sector	Sep 21st, 2021	Energy, emissions, water, environmental impacts	Y	Y	Y (Teng et al., 2023)	/	Y	Subscription
AusLCI	1.42	Australia	Multi-sector	May 1st, 2023	Carbon emissions	Y	Y	N	A1-A3 (ALCAS, 2025)	N	Free
EPiC	2024	Australia	Building materials	Apr 29th, 2024	Embodied GHG emissions, water, energy	Y	N (Crawford et al., 2022)	/	A1-A3	N	Free
ICE	3	Primarily the UK, with some global data	Building materials	Nov 10th, 2019	Embodied carbon	Y	Y	N	A1-A3	N	Free
ÖKOBAUDAT	2023	Primarily Germany	Multi-sector	June 15th, 2023	Energy, emissions, water, environmental impacts	Y	Y	Multiple	A-D	Partly Included	Free

boundaries and included stages, carbon storage accounting methods, and EoL scenarios considered. More than ten materials are considered and extracted, including both the structural and non-structural materials. To calculate the embodied impacts of buildings and ensure comparability, we normalized the functional units to 1 m^3 of product for concrete and timber, and 1 kg for other materials. To convert to a standard unit, the density in profile is used as the first choice to convert. To quantify and assess the variability in GHG emissions at building-level values across databases, relative differences were calculated. The Ecoinvent database was selected as the reference (Takano et al., 2014b; Teng et al., 2023) due to its comprehensive coverage of materials and widespread application. The relative difference of GHG emissions value for each database was calculated using the equation below (Takano et al., 2014b):

$$\text{Relative Difference} = \frac{GHG_{database} - GHG_{reference}}{GHG_{reference}} \times 100\%$$

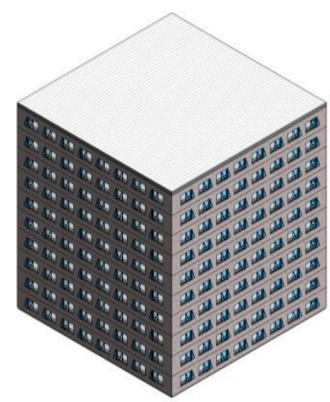
Where GHG_{database} represents the building GHG emissions in the database under evaluation ($kgCO_2eq/m^2$), while $GHG_{\text{reference}}$ represents the building GHG emissions in the Ecoinvent database ($kgCO_2eq/m^2$)

3.3. Case study

To evaluate the practical implications of database selection on embodied emissions calculations, this study analyzes a 10-storey office building with a total gross floor area of 9986 m^2, representing a typical central business district commercial development previously utilized in multiple case studies (Bell et al., 2022; Daly et al., 2014; Samarakoon and Soebarto, 2011). The reference building has a square footprint of 31.6 m × 31.6 m with a 30 % of window to wall ratio (Table 2). Two building variants were modeled to facilitate comprehensive database assessment across different material applications. The RC variant employs a moment-resisting frame system with a 7.9 m × 7.9 m column grid utilizing 400 mm × 400 mm columns and a reinforced concrete core. The floor system comprises a 200 mm structural concrete slab with 60 mm topping layer. To meet structural and load design requirements, the study referenced previous Australian research (Jayalath et al., 2020; Li et al., 2019). The mass timber building incorporates a combination of CLT shear walls and glulam post-and-beam framing, with 160 mm 5-ply CLT floor panels. The design details of the CLT envelope, such as load-bearing wall configurations, follow specifications from the CLT handbook to reflect common industry practices (Swedish Wood, 2019).

Regarding scope boundaries, this analysis focused on the superstructure elements as foundation design is highly dependent on site-specific and geotechnical conditions (Hart et al., 2021). Mass timber structures typically weigh less than concrete alternatives, potentially reducing foundation requirements (Hassan et al., 2019). Incorporating these differences would introduce additional variables beyond the scope of the current investigation. Therefore, the study focused exclusively on superstructures, excluding foundations and below-grade elements. Future research could expand this boundary to include geotechnical elements for more comprehensive mass timber LCA studies (Song et al., 2025).

Table 2
Reference building parameters for embodied emissions analysis.

Parameter	Specification
Visualization	
Typology	10-storey office building
Gross floor area	9986 m^2
Footprint dimensions	31.6 m × 31.6 m
Floor-to-floor height	3.6 m
Window-to-wall ratio	30 %
Analysis scope	Superstructure (excluding foundations)
Functional unit	1 m^2 of gross floor area
Thermal equivalence	Equivalent envelope U-value
Column grid	7.9 m × 7.9 m
Column dimensions	400 mm × 400 mm
Frame system	Moment-resisting RC frame CLT shear walls with glulam post-and-beam framing

4. Results and discussions

4.1. Analysis of differences at the database level

Table 3 shows the availability of data across five databases for

Table 3
Availability of LCA data across five databases for common building materials.

Material	Specific product	AusLCI	EPiC	Ecoinvent	ICE	ÖKOBAUDAT
Wood	Mass timber (CLT, Glulam, etc.)	4	4	13	10	9
	Cork slab, hardwood board, etc.	16	8	178	19	10
Aluminium		3	4	36	18	3
Concrete block		0	3	32	3	5
Cement mortar		1	1	18	14	2
Concrete	Concrete 30 MPa	3	13	25	4	29
	Concrete 50 MPa	3	13	7	3	2
Steel		2	4	20	15	7
PVC		0	2	9	0	4
Insulation	EPS, XPS, mineral wool, etc.	8	6	66	0	25
Particleboard/Plasterboard/Gypsumplaster		4	3	16	6	9
Aggregate		2	3	7	10	10
Glass	Float glass, flat glass	3	3	20	7	8
Membrane and sealing		1	3	22	3	11

common building materials. Ecoinvent consistently demonstrates the most comprehensive coverage, particularly for wood products (178 datasets) and insulation materials (66 datasets), reinforcing its status as a leading LCA database. All databases provide data for mass timber products, with Ecoinvent (13 datasets), ICE (10 datasets) and ÖKOBAUDAT (9 datasets) offering the most variety, reflecting the growing popularity of mass timber in sustainable construction. Traditional structural materials (concrete and steel) demonstrate comprehensive coverage, with all examined databases containing extensive datasets. Insulation materials show significant variability in dataset availability, with Ecoinvent (66 datasets) and ÖKOBAUDAT (25 datasets) offering the most options. However, the analysis revealed significant gaps in the availability of certain material data, such as the absence of concrete block and polyvinyl chloride (PVC) data in AusLCI and the lack of specific insulation datasets in ICE, which could potentially pose challenges for building LCA studies in different regions.

The findings also indicated that different databases have emphasized varied system boundaries and impact assessment methods. For example, ICE focuses on the cradle-to-gate stage (A1-A3), whereas ÖKOBAUDAT provides more than 1200 building product datasets according to EN 15,804 + A1 or EN 15,804 + A2 for multiple stages. Such variation was also observed in the EPiC database, which emphasizes a hybrid approach that combines the process data and input-output data to address the truncation issues of process data for construction materials (Crawford et al., 2022). AusLCI provides several methods about carbon calculation, which include carbon neutrality assumption, total carbon, fossil carbon, biogenic carbon, and infrastructure. The AusLCI website states that all electricity and transport inputs have been adjusted to the Australian context (ALCAS, 2025), but detailed information may be required to ensure accuracy and prevent double-counting or the omission of important stages such as EoL stages.

The accuracy of LCA calculations is often compromised by inconsistencies in system boundaries and impact assessment methods (Anand and Amor, 2017). Moreover, LCA databases do not always provide comprehensive data for primary building materials, let alone secondary materials used in smaller quantities. This is evidenced by the limited datasets for concrete block and PVC in AusLCI database. This often necessitates referring to data from other databases or external references, which may enable the completion of the LCA assessment but introduce other challenges for building LCA. First, it may fail to capture the specific contextual conditions of the project. Second, combining data from multiple sources can create methodological inconsistencies in system boundary definitions (e.g., cradle-to-gate versus cradle-to-grave), background data assumptions, cut-off criteria and also biogenic carbon treatment.

The quantification of environmental impacts requires standardizing functional units to align with the bill of quantities (BoQ). However, not all databases give detailed information about the properties, such as density in AusLCI. In such cases, density values from reference data sources, such as national construction code material properties, are used as substitutes. This conversion will cause disparities and inaccuracies in building GHG emissions assessments. Furthermore, the inability to adjust data to reflect location-specific factors such as energy mix or transport distances further amplifies these challenges. These limitations underscore the need for comprehensive and flexible databases in future development (Teng et al., 2023).

In addition, these databases emphasize and cover different environmental impact indicators. AusLCI and ICE focus solely on carbon emissions, while EPiC provides data on embodied carbon, energy, and water consumption for construction materials. However, these databases neglect other indicators, such as primary energy (renewable and non-renewable), ODP, photochemical ozone creation potential (POCP),

Table 4
Range and variability of embodied GHG values.

Material	Unit	Count	Min	Max	Mean	Median	Std
Aluminium	$kgCO_2eq/kg$	64	1.51	32.70	13.87	12.85	6.97
Concrete block	$kgCO_2eq/kg$	43	0.07	1.20	0.49	0.47	0.34
Cement mortar	$kgCO_2eq/kg$	36	0.10	0.70	0.22	0.17	0.15
Concrete 30 MPa	$kgCO_2eq/m^3$	74	125.90	416.00	236.45	244.59	83.05
Concrete 50 MPa	$kgCO_2eq/m^3$	28	171.82	626.00	418.35	411.50	108.08
Steel	$kgCO_2eq/kg$	48	0.36	5.50	2.38	2.26	1.30
PVC	$kgCO_2eq/kg$	15	1.90	11.20	3.25	2.57	2.18
Mass timber	$kgCO_2eq/m^3$	40	−686.80	1718.00	27.15	147.95	589.80
All other wood	$kgCO_2eq/m^3$	231	−1234.00	3680.00	101.12	69.97	400.93
Insulation	$kgCO_2eq/kg$	105	−1.54	31.11	3.79	3.17	5.10
Particleboard/Plasterboard/Gypsumplaster	$kgCO_2eq/kg$	38	−1.29	1.48	0.29	0.43	0.65
Aggregate	$kgCO_2eq/kg$	32	0.00	0.39	0.04	0.01	0.08
Glass	$kgCO_2eq/kg$	41	0.17	3.11	1.56	1.63	0.68
Membrane and sealing	$kgCO_2eq/kg$	40	0.19	6.40	1.55	0.90	1.64

Note: GHG emissions values are compiled from the five LCA databases analyzed in this study (Ecoinvent, AusLCI, EPiC, ICE, and ÖKOBAUDAT).

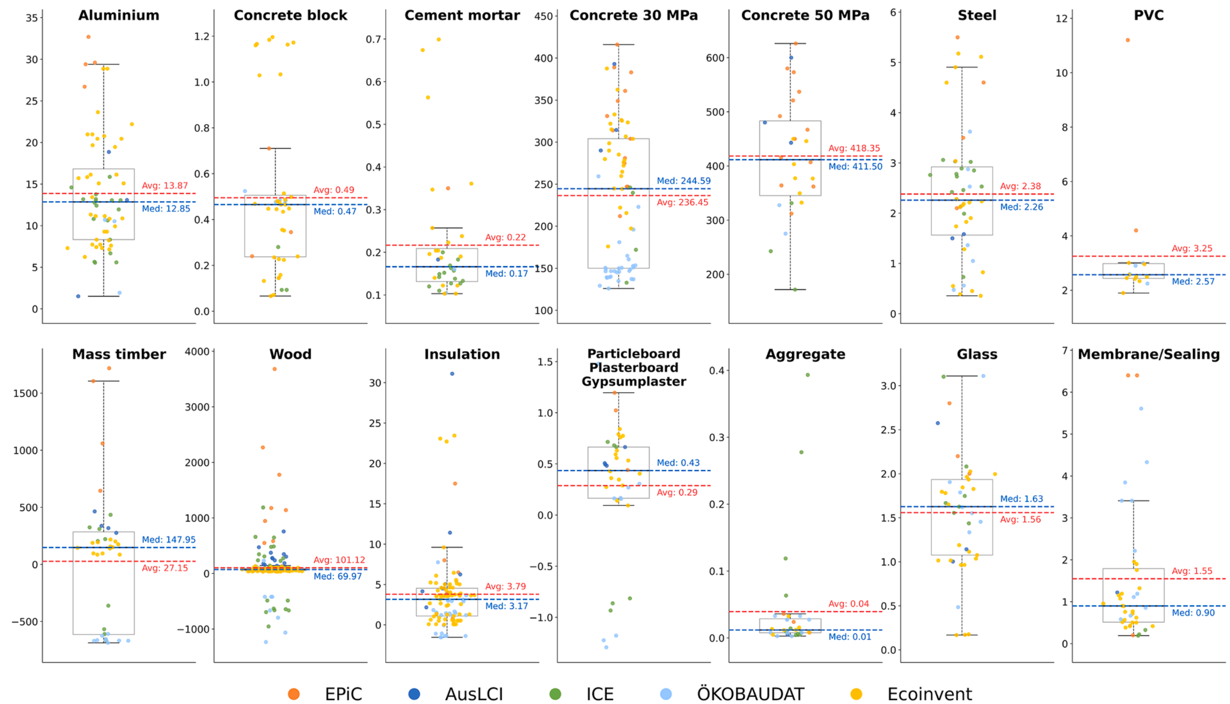


Fig. 1. GHG emissions factors for construction materials ($kgCO_2eq/m^3$ for timber and concrete; $kgCO_2eq/kg$ for other materials).

and eutrophication potential (EP). Such omissions hinder a broader perspective on sustainability, as GWP is not the only concern. Additional impact indicators are also essential to provide a more comprehensive understanding of research and public decision-making. In contrast, Ecoinvent and ÖKOBAUDAT perform well in terms of the comprehensiveness of indicators.

4.2. Analysis of differences at the material level

The analysis of the statistical summary (Table 4) and the box plots (Fig. 1) reveals the variability in GHG emissions data across different building materials across the analyzed databases. This analysis focuses exclusively on production stage (A1-A3) emissions where detailed life cycle stage information is available. Regarding the values, the Ecoinvent data employs the IPCC2013 methodology, while AusLCI datasets utilize the carbon neutrality assumption approach. The analysis revealed that the maximum GHG emissions are approximately 21.7 times and 15.5 times higher than minimums for aluminum and steel, respectively. This finding aligns with previous research (Pomponi and Moncaster, 2018), which reported substantial variations in embodied carbon coefficients between different materials (284–1044 %). According to Fig. 1, in general, the EPiC database tends to report higher emissions values, likely due to its hybrid methodology combining process-based and input-output data, which introduces broader system boundaries but amplifies uncertainty. In contrast, ÖKOBAUDAT and ICE often show more conservative estimates, with some bio-based materials showing negative values due to carbon storage. The most pronounced variation is observed in wood products, where emissions exhibit both negative and positive values, spanning a comprehensive range of 4914 $kgCO_2eq/m^3$. As shown, the EPiC database consistently reported the highest values, while ÖKOBAUDAT reported negative values, highlighting fundamental differences in carbon sequestration accounting methodologies. Outliers exist in concrete blocks, cement mortar, steel, PVC, insulation and wood. This raises concerns about the data quality issues for LCA application, including inconsistent system boundaries (e.g., inclusion/exclusion of transportation, Module D), regional data gaps, poor transparency in allocation methods, and timeliness issues of datasets. For instance, outdated manufacturing data may misrepresent current decarbonization efforts in steel and concrete production, while lack of dynamic biogenic carbon modeling undermines the credibility of negative emissions claims for mass timber.

4.2.1. Conventional building materials

Aluminum, which is commonly used for window frames, exhibits wide emission ranges from 1.51 to 32.70 $kgCO_2eq/kg$. The box plot reveals that ÖKOBAUDAT shows a more concentrated distribution around the median, while EPiC database reports generally higher values and outliers (26.70–32.70 $kgCO_2eq/kg$). This variation is also attributed to inconsistent approaches in accounting for recycled content and energy sources. Some databases, like AusLCI, show primary aluminum at 18.86 $kgCO_2eq/kg$ versus second aluminum at merely 1.51 $kgCO_2eq/kg$. This variation reflects a great methodological challenge in recycling accounting. If the product is characterized as secondary metal from old scrap (an upstream flow from previous life cycles) at the production stage (A1-A3) and recycling credits are considered (a downstream flow to future life cycles), there is a potential risk for double counting of environmental benefits. This overlap between upstream recycled content and downstream recycling potential raises concerns in terms of material flow and the system boundary in recycling calculations especially for those widely recycling metals.

Steel in EPiC reports the highest range of values (2.10–5.50 $kgCO_2eq/kg$), while ICE displays a more concentrated distribution around the median (2.26 $kgCO_2eq/kg$). The box plot reveals several outliers in the Ecoinvent and EPiC databases, likely due to variations in production processes or system boundary definitions, such as converter versus electric arc furnace methods and differences in the percentage of scrap content used.

In addition, the disparities may be attributed to the electricity carbon intensity in the source country/location for the manufacturing process, highlighting the importance of spatial specificity in LCA studies. For instance, data from the ICE database shows that aluminum emissions in North America and Europe are approximately 5.58–5.65 $kgCfO_2eq/kg$, while in China and Africa, the values can range from 12.40–14.59 $kgCO_2eq/kg$. These regional differences have significant implications for both research methodology and policy development. From a research perspective, they underscore the urgent need to develop regionalized carbon coefficients that accurately reflect local energy mixes and manufacturing practices, particularly in regions with limited database availability. The significance of such implications demonstrated in the findings in Kyaw et al. (2024), which showed up to 95 % reduction in embodied emissions when transitioning from generic global data to site-specific environmental impact assessments. Spatially explicit LCA, which integrates geographical information into life cycle inventories (LCIs) and/or life cycle impact assessments (LCIAs), can enhance the accuracy of environmental impact quantification (Shi and Yan, 2024). From a policy standpoint, these variations present substantial challenges for standardizing embodied impact benchmarks across different geographical contexts. Consequently, using data from regions with different energy mixes and manufacturing technologies without contextual adjustments can lead to significant underestimation or overestimation of environmental impacts.

The analysis of 30 MPa concrete demonstrates substantial variation (125.90 to 416.00 $kgCO_2eq/m^3$, Std = 83.05), while 50 MPa concrete shows even greater disparity (171.82 to 626.00 $kgCO_2eq/m^3$, Std = 108.08). Concrete production exhibits substantial variation depending on the proportion of supplementary cementitious materials, with ground granulated blast furnace slag (GGBFS) or fly ash content significantly influencing embodied carbon emissions.

Cement-based materials (concrete blocks and mortar) generally show more consistent emissions patterns across databases (Std = 0.34 and 0.15 respectively). However, notable outliers exist, particularly in the Ecoinvent database for concrete blocks (1.20 $kgCO_2eq/kg$) and cement mortar (0.70 $kgCO_2eq/kg$). The GHG emissions for PVC ranged from 1.90–11.20 $kgCO_2eq/kg$, with EPiC showing higher values (4.20–11.20 $kgCO_2eq/kg$) compared to other databases. Glass emissions data presents moderate variation (0.17–3.11 $kgCO_2eq/kg$). Some of the variations reflect the different glass types (float, toughened, laminated) or production technologies. The emissions data for board materials shows a relatively narrow range (−1.29 to 1.48 $kgCO_2eq/kg$). The variation between databases, though smaller than other materials, highlights the importance of clear product category definitions.

The analysis of insulation materials reveals great disparities in embodied carbon reporting across construction materials. The insulation materials include Expanded Polystyrene (EPS), Extruded Polystyrene (XPS), rockwool, wood fiber insulation, etc. The analysis shows an extraordinary range from −1.54 to 31.11 $kgCO_2eq/kg$, with several notable patterns and extreme outliers that warrant careful use. The outliers mainly come from Ecoinvent, but the highest values come from XPS, HFC-134a blown (31.11 $kgCO_2eq/kg$) in AusLCI. This is attributed to the type of blowing agents used in foam insulation production (Teng et al., 2023). HFC-134a blowing agent has a GWP100 value of 1370 (Shrestha et al., 2014). The other extreme is from ÖKOBAUDAT database with negative values of −1.54 $kgCO_2eq/kg$ for wood fiber insulation (German average), which suggests potential carbon sequestration. EPiC database values show a wider spread (0.79 to 8.00 $kgCO_2eq/kg$) but remain more conservative compared to the extremes.

4.2.2. Bio-based materials: mass timber

Mass timber exhibits the most extreme variability than any other materials, with emissions ranging from −686.80 to 1718.00 $kgCO_2eq/m^3$ (mean = 27.15, Std = 589.80). According to the review (Younis and Dodoo, 2022), the GWP of 1 m^3 of CLT (A1-A3) averages 152.0

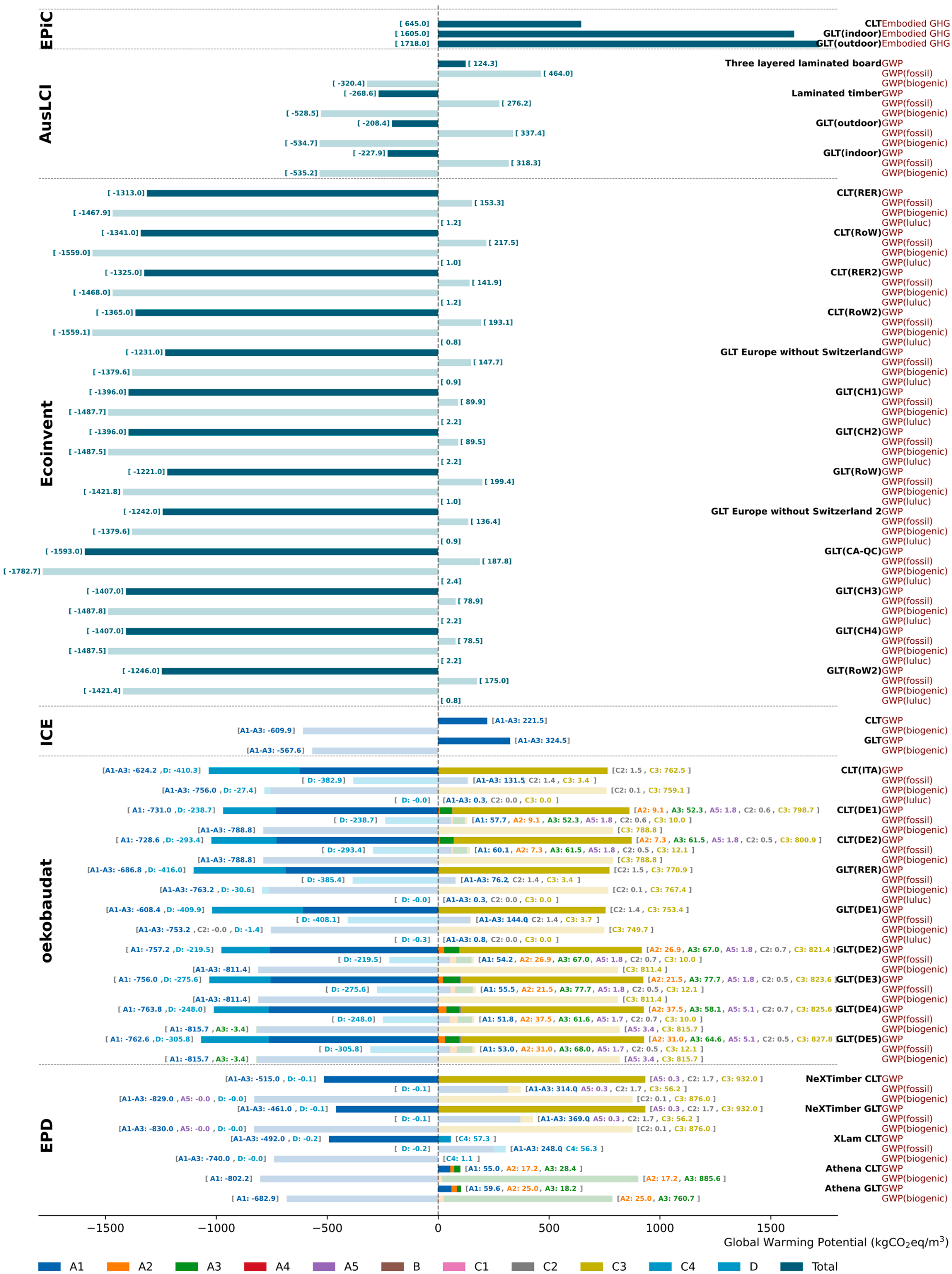


Fig. 2. Detailed emissions breakdown for mass timber products across different data sources.

$kgCO_2eq/m^3$ with a higher Std of 118.4. This can be attributed to the variability in the energy sources and technology among different manufacturers in different regions (Rasmussen et al., 2021).

This extreme range underscores the methodological inconsistencies in accounting for biogenic carbon and system boundaries in mass timber datasets. To better illustrate, Fig. 2 gives detailed emissions of mass timber across diverse data sources. For datasets with multiple EoL scenarios, only the default or first EoL scenario is represented and shown. This significant heterogeneity in emissions data, if not well addressed, will have great implications for the reliability and comparability of mass timber LCA studies. Similarly, other wood products display substantial variability, though with less extreme outliers to mass timber. Outliers in timber materials mainly come from the EPiC databases. Specifically, the embodied GHG emissions for CLT are 645 $kgCO_2eq/m^3$ while values are 1718 for indoor glulam and 1605 for outdoor glulam. These extremely high values are related to the hybrid methods.

For 1 m^3 of CLT in EPiC, the process-based GHG emissions are 377 $kgCO_2eq$, while input-output calculations yield 598.80 $kgCO_2eq$, resulting in a value of 644.70 $kgCO_2eq$ using the hybrid method. The proportional contribution in the hybrid method comprises 46 % process-based and 54 % input-output data. However, for Glulam products, the distribution is quite different. Indoor Glulam exhibits process-based emissions of 377 $kgCO_2eq/m^3$, with substantially higher input-output values of 2348 $kgCO_2eq/m^3$, resulting in a hybrid value of 1718 $kgCO_2eq/m^3$. Similarly, outdoor Glulam shows process-based emissions of 397.80, input-output values of 2226, and a hybrid result of 1605 $kgCO_2eq/m^3$. This pattern aligns with findings reported by Venkatraj and Dixit (2021), who observed that hybrid input-output calculations yield higher embodied energy values compared to process-based assessments. This difference can be attributed to the incorporation of macro-economic data to cover broader system boundaries (Venkatraj and Dixit, 2021).

A notable variability in the proportion of process data exists within hybrid coefficients in EPiC. Timber has a relatively lower proportion of process-based data, accounting for only 18–20 % of the total hybrid value for Glulam products. This finding aligns with Lausselet et al. (2022), who reported similar patterns for timber products (25 % process-based data for softwood and 39 % for indoor glulam) in their analysis of hybrid EPiC coefficients in a Norwegian context, further highlighting the significant contribution of input-output data in timber assessments. This disparity arises from usage of coal along the value chains (Lausselet et al., 2022) and significant contribution of transportation emissions (Crawford et al., 2022; Lausselet et al., 2022). Transportation data are predominantly calculated using input-output methodology due to limitations in process-based transport data reliability and comprehensiveness (Crawford et al., 2022). Such impacts are notably significant, with road transport contributing 14.0 % of total GHG emissions for Glulam (the highest single contributor) and 9.6 % for CLT (the second-highest contributor).

The input-output data enhances coverage of stages to address the truncate issue; however it also introduces potential limitations in accuracy and uncertainties (Wiedmann, 2009; Wiedmann et al., 2011). This is primarily because input-output data relies on national averages, which may not accurately capture the specific characteristics and conditions of materials subject to analysis (Chang et al., 2012). Especially in context of globalized supply chains, the production and delivery of individual products result in distributed resource consumption and emissions across multiple geographical regions (Shi and Yan, 2024). This becomes particularly relevant as mass timber serves as the main structural material. The transportation distances and processing methods of mass timber can vary significantly across different geographical contexts. These regional variations in timber supply chains may not be adequately represented in the input-output approach.

The findings of this study reveal additional complexities in the examination of underlying data sources. Specifically, the process-based values in EPiC utilize AusLCI profiles, employing "Glued laminated timber, indoor use, at plant" for CLT and indoor Glulam, and "Glued laminated timber, outdoor use, at plant" for outdoor Glulam. Since no exact CLT product exists within the AusLCI database, the use of Glulam profiles as a reference for the CLT dataset may introduce inaccuracies in emissions assessments.

This study revealed that data timeliness is a critical issue, which can potentially affect the accuracy and relevance of LCA assessments. Process-based emissions values in EPiC (377 $kgCO_2eq/m^3$ for indoor and 397.80 $kgCO_2eq/m^3$ for outdoor glulam) show significant differences from AusLCI values (318.34 $kgCO_2eq/m^3$ and 337.42 $kgCO_2eq/m^3$ respectively). This disparity may stem from version differences, as EPiC references AusLCI version 2018.06.19, whereas the studied AusLCI database (version 1.42, released May 2023) contains updated values. Moreover, for input-output data, significant time gaps often exist between the reference year of underlying data and current policy and research questions that aimed at (Wiedmann et al., 2011). For instance, research conducted in 2016 utilized input-output data from 2010 (Guan et al., 2016), while 2007 economic data was applied in 2012 study (Chang et al., 2012). This delay may occur due to the large volume of data collection and processing procedures for national economic accounts. However, these temporal considerations may become particularly relevant for mass timber, which is characterized by rapid technological advancements, evolving forestry practices, and shifting global trade patterns. By addressing these temporal considerations, researchers can enhance the reliability and relevance of LCA results for timber products.

A significant limitation in mass timber LCA emerges from the methodological variations across different impact assessment methods for GWP calculations. For instance, within the Ecoinvent database, IPCC2013 and EF v3.0 yield similar results for CLT (193.72 and 194.94 $kgCO_2eq$, respectively), while EF v3.0 EN15804 presents quite different results, with a total climate change impact of −1365.15 $kgCO_2eq$. This disparity primarily arises from differences in biogenic carbon accounting methods, with EF v3.0 EN15804 providing a more detailed breakdown including biogenic (−1559.11 $kgCO_2eq$), fossil (193.14 $kgCO_2eq$), and land use change (0.82 $kgCO_2eq$) components. The fossil emissions are similar to the other methods. The treatment of other GWP indicators creates significant disparities in total emissions values.

Furthermore, disparities exist even when comparing methods regarding biogenic carbon flow. For example, the CLT profile from Athena EPD focusing on the North American context (Athena Sustainable Materials Institute, 2021) reports GWP values with biogenic carbon for A1-A3 of −802.24, 17.24, and 885.57 $kgCO_2eq$, respectively. The report indicates biogenic carbon release in the A3 stage. This differs from the EN 15,804 methodology, where biogenic carbon release typically occurs in the EoL stage after demolition. Such methodological differences require careful consideration when using EPDs, as they may not fully align with the standards used in generic databases. These different modeling approaches result in significant variations in both results and carbon flow patterns within the system. In addition, the land-use and land-use-change (LULUC) GWP is not identified in most data.

Further complexity arises from variations in material specifications and manufacturing processes. The environmental impact of mass timber is significantly influenced by adhesive selection, with different options such as Melamine-Urea-Formaldehyde (MUF), Polyurethane (PUR), or average glue mix in Glulam yielding varying environmental profiles in Ecoinvent. In addition, several EPDs distinguish different profiles for preservative treatments or untreated. For NeXTimber (Timberlink Australia, 2023), sawn timber is treated (H3 Propiconazole + Tebuconazole Light Organic Solvent Preservatives (LOSP)), and for XLam (XLam, 2021), the optional treatment is Hyne T3 Plus. Specifically, Hyne T3 Plus in XLam contributed 11.4 kg more GWP, and H3 treatment contributed 61 kg more fossil GWP for GLT in NeXTimber for a reference flow of 1 m^3. While in EPD System Red Stag Wood Solutions CLT (Red Stag Timber, 2022), it provides multiple datasets for different treatments with slightly different GWP. These datasets include no treatment, H1.2 Boron, H3.1 LOSP, H3.1 Copper Azole, H3 Copperchrome-arsenate (CCA), H4 CCA, and H4 CCA Re-dried. It also emphasized that some scenarios are not applicable for energy recovery.

Current LCA databases often oversimplify material specifications, which can introduce significant uncertainties in building-level LCA results. These limitations in material specification present a great challenge for researchers who may lack detailed information about the manufacturing processes, making it difficult to match products that truly reflect the materials being used in construction. This uncertainty becomes particularly significant for main structural materials like timber and concrete. To address these challenges, future LCA databases should incorporate enhanced transparency of material specifications and manufacturing processes, such as clearly indicating the type of adhesives utilized in mass timber, specific mix designs in concrete and recycled content in steel.

The implementation of a comprehensive documentation system at the construction stage is essential to capture detailed material profiles. While these data are easily accessible at the beginning stages, they become extremely difficult to gather when conducting LCA afterward. Therefore, creating a detailed material passport becomes particularly vital for all buildings, especially those with mass timber elements. This documentation system could integrate with BIM to track key parameters such as building design, material profiles, quantity usage, and on-site waste generation. This can help researchers to conduct more reliable LCAs that reflect context-specific situations. Moreover, the establishment of a standardized documentation framework ensures that construction data serves as a reliable reference for future building maintenance, renovations, disassembly and EoL scenarios. Future policies could introduce incentives to encourage the adoption of material passports and integrate into green building certification systems, such as LEED, Green Star (Australia), and BREEAM.

Another critical issue in accurately calculating biogenic carbon lies in modeling the biogenic carbon release after demolition. Darby et al. (2013) have shown substantial disparities across different EoL scenarios (reuse, reengineer, incinerate, energy recovery and landfill). However, current databases show limited coverage of recycling scenarios, with only ÖKOBAUDAT providing some data for certain scenarios (thermal recovery and material reuse). EPDs generally offer more detailed and diverse EoL scenarios that are compared in the analysis, but they also reveal significant disparities in their data. This inconsistency underscores the need for databases to include more reliable data on recycling and reuse stages, as well as careful utilization of these data.

While some EPDs consider landfills as the primary disposal method, others focus on energy recovery or implement mixed scenarios combining energy recovery with recycling (Financiera Maderera, 2023). For instance, in Europe, incineration is often the default treatment method, as demonstrated in the Stora Enso CLT EPD (Stora Enso Oyj, 2023). This approach, involving timber dismantling and chipping for thermal processing, offsets conventional heat production from natural gas and high-voltage electricity generation. These varying EoL strategies reflect diverse policies and preferences for mass timber products, influenced by geographical and regional differences.

The accounting of Module D credits presents a significant methodological challenge, particularly regarding the allocation of benefits between current and future life cycles. Rasmussen et al. (2021) argued that Module D should be excluded from the product system as it belongs to the next life cycle, primarily to avoid double counting of environmental benefits. This approach may explain some datasets from AusLCI (e.g., primary aluminum at 18.86 $kgCO_2eq/kg$ versus secondary aluminum at 1.51 $kgCO_2eq/kg$). This may address double counting but may introduce another challenge. For example, in cases where timber undergoes energy recovery through incineration, excluding Module D benefits from the current assessment may underestimate the environmental advantages since no subsequent building life cycle exists for the bio-based material. Moreover, this approach complicates cross-sectoral environmental accounting, particularly when benefits span multiple sectors such as buildings, energy infrastructure, and forestry management (Andersen et al., 2024c). Furthermore, excluding Module D will inadvertently discourage design strategies that promote material circularity. Design for disassembly and other efforts to improve the quantity and quality of recoverable mass timber may appear less impactful or not immediately apparent if their benefits are not reflected.

The inclusion of Module D in the current life cycle also introduces methodological challenges regarding benefit allocation and circular economy principles. In datasets from ÖKOBAUDAT, reuse scenarios contribute to an 11.16–13.30 $kgCO_2eq$ fossil fuel saving, while incineration contributes to 219.50–416.00 $kgCO_2eq$ savings. This creates a potential misalignment between environmental assessment results and circular economy principles. The cascading principle, which prioritizes reuse over incineration, should always take precedence to enhance material efficiency, preserve biogenic carbon storage, and achieve long-term environmental and economic benefits. This approach reflects a fundamental distinction in biogenic carbon truncation: reuse scenarios maintain the carbon storage function and transfer material value to subsequent building life cycle, while incineration terminates the life cycle for immediate energy benefits, permanently releasing stored carbon.

For 1 m^3 of mass timber, the savings from reuse scenarios range from 351 to 447 $kgCO_2eq$ in certain Australian EPDs (Timberlink Australia, 2023; XLam, 2021), which differs significantly from the ÖKOBAUDAT datasets. This may relate to the baseline fossil fuel GWP in the product stage but concerns also exist regarding the quality of EPD data. For instance, in (XLam, 2021), the 447 $kgCO_2eq$ saving surpasses the A1–A3 stage fossil fuel GWP of 248 $kgCO_2eq$, raising questions about data accuracy. In (Timberlink Australia, 2023), biogenic savings of −43.2 $kgCO_2eq$ in the reuse D stage appear unexplained. Most notably, the KLH CLT EPD (KLH Massivholz GmbH, 2023), includes biogenic carbon storage in the reuse stage as credits, showing a total of −852 $kgCO_2eq$ GWP for module D. This figure comprises 87.6 $kgCO_2eq$ in fossil fuel savings, 762 $kgCO_2eq$ in biogenic carbon credits, and 2.09 $kgCO_2eq$ in LULUC GWP. This approach contrasts with other EPDs and datasets. These methodological inconsistencies raise significant concerns about data reliability and comparability when quantifying building emissions.

Several key considerations regarding mass timber require attention in building LCA studies. First, the prefabrication process requires special consideration in the construction stage. Takano et al. (2015) proposed splitting the construction process stage into two distinct modules: A4–5:P (prefabrication) and A4–5:O (on-site). The A4–5:P module encompasses transportation from the factory gate to the prefabrication facility and all associated manufacturing processes. This division enhances result transparency and enables better tracking of waste flows during prefabrication. However, this detailed process breakdown is often missing in databases, raising concerns about the standardization of life cycle modules for mass timber assessment. Furthermore, packaging requirements become particularly needed, especially when mass timber must be transported over long distances. Protective materials such as wrapping, timber spacers, and additional weatherproofing may be needed to prevent damage during extended transport and storage periods. In Xlam CLT EPD (XLam, 2021), low-density polyethylene wrap was used to cover CLT from exposure to the elements, and CLT is placed on softwood timber gluts and bearers during the transport and storage on site. Specifically, packaging contributed to 10.5 kg fossil GWP (∼4.23 % of the cradle to gate value) and −6.3 kg of biogenic emissions for 1 m^3 of CLT. In some datasets from ÖKOBAUDAT, the A5 stage exclusively contains the disposal of packaging materials, but it is not always present in other datasets. These auxiliary materials should be considered in data to ensure a holistic assessment, especially for those databases that have limited transparency.

In addition, these methodological challenges in LCA apply not only to mass timber but also to other bio-based materials such as bamboo. This is particularly relevant for regions like China that have limited

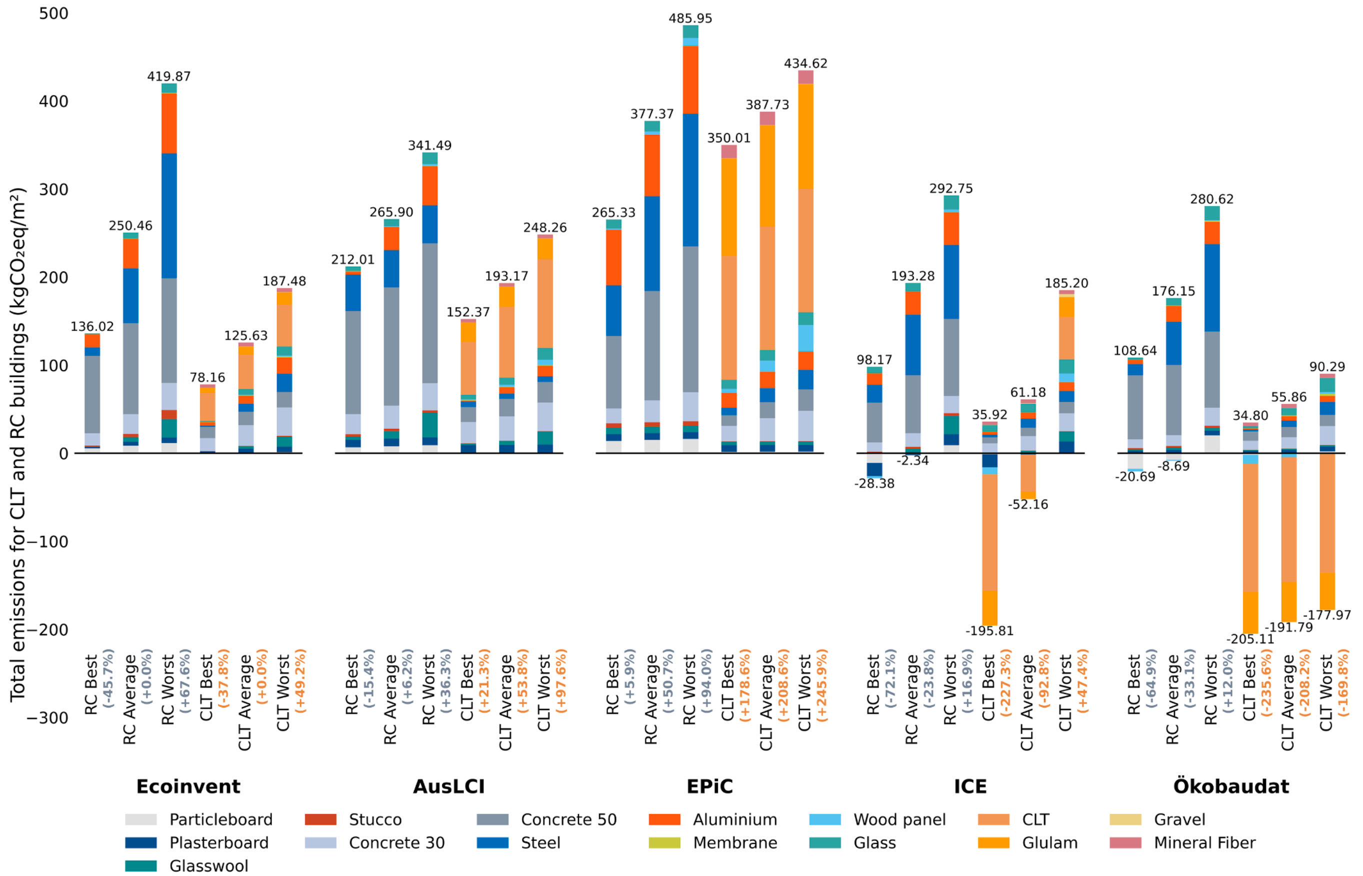


Fig. 3. Building level emissions from different databases for the product stage (A1-A3).

forest resources for timber production but possess abundant bamboo resources. For example, Liu et al. (2024) evaluated structural glued laminated bamboo and reported that the product stores more than twice the amount of biogenic carbon (1140 $kgCO_2eq/m^3$) compared to the emissions during its production (467 $\pm$ 9.1 $kgCO_2eq/m^3$).

4.3. Analysis of differences at the building level

The analysis of product stage emissions (Fig. 3) presents the best, average, and worst scenarios for mass timber buildings for each database, with average Ecoinvent values serving as the baseline. While the material-level analysis does not distinguish between material types, the building-level analysis accounts for specific materials like insulation and sealing. When certain materials like bitumen sealing are missing from databases, similar materials such as asphalt are substituted. Some databases contain limited data resources. For instance, the single CLT dataset in EPiC may not adequately represent the range of scenarios or regional variations. Even more concerning, some databases may completely lack certain material categories. For missing elements like insulation in ICE, Ecoinvent data is used to maintain consistency without introducing additional uncertainty.

RC buildings show emissions ranging from 69.79 to 485.95 $kgCO_2eq/m^2$, while mass timber buildings display wider variation from -170.31 to 434.62 $kgCO_2eq/m^2$. The results are compared with several studies to validate these findings. For instance, a study carried out in US reported embodied emissions (A1-A5) of about 450 $kgCO_2eq/m^2$ for RC and 327–330 $kgCO_2eq/m^2$ for CLT buildings (Pierobon et al., 2019). Robati and Oldfield (2022) found embodied emissions of mass timber buildings ranging between 196–590 $kgCO_2eq/m^2$, averaging 417 $kgCO_2eq/m^2$ for mass timber constructions. For concrete buildings, emissions ranged from 307–618 $kgCO_2eq/m^2$, with an average of 465 $kgCO_2eq/m^2$. While these studies provide valuable references, comparisons across studies present challenges due to differences in scope, architectural and structural design, and material emission factors. Helal et al. (2020) demonstrated significant variability in embodied carbon assessments (A1-A5), revealing up to 22 % variation attributable to differences in design methods and structural loads. System boundaries present another significant challenge in quantifying embodied impacts. Specifically, Robati and Oldfield (2022) included stages A1-A5, B1, B4, C1, C3, and C4 in their analysis, while Li et al. (2019) focused solely on cradle-to-gate (A1-A3) emissions. These variations in scope and modules may be driven by data availability and differences in the definition of system boundaries.

RC buildings consistently show positive emissions across all databases, with concrete and steel being the primary contributors. Concrete and steel consistently emerge as the predominant contributors to embodied carbon emissions across all analyzed scenarios in the five databases, collectively accounting for an average of 78.29 % of the total embodied carbon footprint. Aluminum emissions in RC buildings vary significantly, ranging from 3.55 $kgCO_2eq/m^2$ (AusLCI best case) to 76.98 $kgCO_2eq/m^2$ (EPiC worst case). Steel emissions show similar variation, spanning from 9.73 $kgCO_2eq/m^2$ (Ecoinvent best case) to 150.67 $kgCO_2eq/m^2$ (EPiC worst case). These variations reflect differences in manufacturing processes, recycled content allocation, system boundaries, and methods.

For mass timber construction, CLT material emissions range from -145.78 to 140.42 $kgCO_2eq/m^2$. ICE and Ökobaudat databases show negative values due to biogenic carbon accounting. CLT buildings typically have lower emissions because they incorporate more bio-based materials in building envelopes and use less carbon-intensive materials like steel and concrete. However, the EPiC database consistently shows higher positive values for both CLT and Glulam. Specifically, CLT and

Glulam contribute 140.42 and 114.91 $kgCO_2eq/m^2$, respectively, in the average scenario in EPiC, accounting for 36.22 % and 29.64 % of total embodied GHG emissions. This suggests that timber construction may not always offer significant carbon reduction benefits. In addition, increasing timber in buildings raises challenges due to implications for harvest and forest management. As emphasized in the literature (Maierhofer et al., 2024; Soimakallio et al., 2022), the unrealized forest carbon storage potential due to harvest is widely overlooked. Specifically, Maierhofer et al. (2024) reported that in none of the case study scenarios were the GHG substitution effects of using wood able to compensate for the forest carbon storage potential foregone due to harvest activities. This finding emphasizes the importance of optimizing timber usage through enhanced circularity principles. Rather than pursuing maximum timber content in buildings, the construction sector should prioritize rational material use that balances carbon benefits with forest preservation and shifts to long-lived construction materials (Arehart et al., 2021). This necessitates a shift from viewing timber as a straightforward low-carbon alternative to understanding it as a finite resource requiring careful life cycle optimization. Recent research analyzing 45 real building LCAs (Andersen et al., 2024a) also found that the amount of timber materials used has only a weak and negative correlation with GWP. However, the study revealed strong correlations between plastic, insulation, and composite materials and other indicators such as photochemical ozone creation, ODP, and EP. This underscores the need to consider and optimize both timber usage and other building materials while considering multiple environmental impacts in mass timber construction design.

Best-to-worst case ranges show significant uncertainty in calculations. In worst-case scenarios across databases, cement-based materials and metals like steel and aluminum are the largest contributors to emissions. While wood panels show extremely high carbon intensity (3680 and 1118 $kgCO_2eq/kg$ in EPiC and ICE, respectively), their relatively small volume compared to concrete and steel results in minimal contribution to RC building emissions. The findings also showed that aluminum consistently presents significant emissions when used.

Within the Ecoinvent database, comparative analysis reveals substantial intra-database variability, with worst-case RC scenarios exhibiting emission values more than three times higher than corresponding best-case scenarios. This pronounced differential underscores the significant methodological uncertainties inherent in building-scale LCA calculations. The AusLCI database demonstrates emission profiles that generally align with Ecoinvent's distributional patterns, though with noticeably reduced internal variance. Specifically, RC buildings exhibit values spanning from 212.01 to 341.49 $kgCO_2eq/m^2$, while mass timber constructions demonstrate a comparable absolute range but at lower overall magnitudes (152.37–248.26 $kgCO_2eq/m^2$).

EPiC database shows the highest values across all scenarios, with worst-case emissions reaching 485.95 $kgCO_2eq/m^2$ for RC buildings and 434.62 $kgCO_2eq/m^2$ for CLT buildings. The high CLT building emissions (350–435 $kgCO_2eq/m^2$) result from higher emission intensities for CLT and Glulam materials combined with larger quantities of mass timber usage, contributing 59.65–71.83 % of the total building footprint in the EPiC scenarios.

The results also show that ICE and ÖKOBAUDAT databases present distinct emission profiles, with CLT buildings showing predominantly negative values (reaching −170.31 $kgCO_2eq/m^2$ in ÖKOBAUDAT) while RC buildings maintain moderate emissions compared to other databases. While EPiC consistently shows the highest concrete and steel emissions, ICE and ÖKOBAUDAT databases report lower concrete emissions. Treatment of biogenic carbon explains negative values in CLT buildings. This underscores the importance of modeling biogenic carbon release in EoL scenarios for timber and other bio-based materials to avoid overestimating carbon capture benefits.

The challenges associated with biogenic carbon flow for bio-based materials are particularly evident at the building level, where no standardized or universally accepted method exists. In fact, the methods varied among countries and have been discussed in (Hoxha et al., 2020; Ouellet-Plamondon et al., 2023). Specifically, Ouellet-Plamondon et al. (2023) discussed the 0/0, −1/1, and −1/1* methods on a multistorey residential timber buildings from 16 countries. The −1/1 method accounts for biogenic carbon flow by calculating both carbon uptake during production and release after demolition. This method assumes that the bio-based material is sourced from sustainably managed forests with continuous rotation, thereby maintaining a constant level of carbon in the forest carbon pool to be considered carbon neutral (Andersen et al., 2024b). Both 0/0 and −1/1 methods only influence the relative contribution of each life cycle stage, rather than the total carbon footprint, as they aim for net zero biogenic carbon. The −1/1* method offers a different approach that considers EoL scenarios of bio-based materials, which may result in a not net zero biogenic balance. In addition, some previous studies (Duan et al., 2022; Jayalath et al., 2020), excluded biogenic carbon calculations but modeled avoided GHG emissions when recovered bioenergy replaces fossil fuel. This approach aligns more closely with a modified −1/1* method and may resemble 0/0* in practice. Regional differences in landfill policies further complicate this EoL modeling. For instance, landfilling wood is prohibited in Sweden (Dodoo et al., 2014a; Elginoz et al., 2024) because it may generate methane, which has a stronger effect on global warming compared to CO_2. In contrast, some regions considered treating landfilled wood as partially or fully sequestered carbon (Ouellet-Plamondon et al., 2023).

This biogenic carbon issue, when combined with the system boundary, introduces an additional layer of complexity for quantifying emissions and comparing results. The assessment scope, particularly when limited to the product stage and applying −1/+1 or −1/+1* carbon accounting methods, can yield negative emissions values. In contrast, studies that exclude biogenic carbon calculations in the product stage often report different outcomes. This variation becomes evident in CLT material emissions data, which ranges from −145.78 $kgCO_2eq/m^2$ (ÖKOBAUDAT) to 140.42 $kgCO_2eq/m^2$ (EPiC). Glulam products demonstrate similar patterns, with emissions varying from −47.50 to 118.82 $kgCO_2eq/m^2$. According to Rasmussen et al. (2021), 18 EPDs account for biogenic carbon uptake while omitting the release phase. This incomplete accounting can result in negative GWP values for timber products and potentially overstate their climate mitigation benefits. In the analysis of a hypothetical 43-story building, Li et al. (2019) examined embodied emissions (A1-A3) across three scenarios: concrete construction (195 $kgCO_2eq/m^2$), timber non-structural elements (80 $kgCO_2eq/m^2$), and timber with a concrete core (−311 $kgCO_2eq/m^2$). This underscores the need for consistent system boundaries in LCA across different building materials. Focusing solely on the product stage may underestimate the environmental impacts of timber buildings. A more comprehensive evaluation, including EoL scenarios in embodied impacts such as carbon release or recycling, provides a clearer picture of their overall carbon flow. However, this raises concerns about uncertainty at the EoL stage, as future recycling processes may vary significantly due to technological advancements.

Quantifying embodied impacts requires consideration not only of material-specific emission coefficients but also of the accuracy of building material quantities. These quantities can be influenced by BIM models, on-site material waste, and other factors. For example, BIM models often simplify complex building structures, particularly timber buildings, due to the intricate nature of their connections (Nawrocka et al., 2023). The level of development (LOD) in BIM models significantly affects embodied emissions (Kyaw et al., 2024; Nawrocka et al., 2023). This is reflected in the findings of Nawrocka et al. (2023) who found that less detailed models (LOD 200) yielded 14.7 % lower embodied impacts compared to more detailed models (LOD 350/400). Similarly, in a Norwegian context, using LOD 350 instead of LOD 200 resulted in a 142 % increase in GWP (Kyaw et al., 2024). The limitations of using BIM models to derive material quantities have been increasingly recognized, particularly due to the absence of empirical validation against 'as-built' material quantity inventories. To address this limitation, D'Amico and Arehart (2024) proposed the data-driven probabilistic graphical method as an alternative approach to estimate material quantities and embodied carbon in building structures. Future research might potentially benefit from integrating such data-driven methodologies to enhance the accuracy of material quantity estimations. Conducting detailed LCA requires significant effort to ensure accuracy in both carbon coefficients and building material quantities. The quality of building material quantities and material environmental datasets shapes the calculation of embodied impacts. Using site-specific environmental data rather than generic global data significantly improves the reliability and accuracy of LCA results. Studies have demonstrated variations ranging from 15 % differences in GWP between product-specific EPDs and generic databases (Loli et al., 2023) to up to 95 % reductions when transitioning from generic to site-specific environmental data (Kyaw et al., 2024). This underlines the importance of developing localized databases for building materials that accurately reflect regional manufacturing processes, local energy mix (e.g., renewable or fossil-based), and consistent LCA methodologies. Further work is needed to standardize LCA methodologies, calculation approaches, and system boundaries across the industry.

Datasets from LCA databases often emphasize the cradle-to-gate stage, specifically covering material extraction and production (A1-A3). This methodological emphasis on initial life cycle stages aligns with Li et al. (2019), who specifically examine embodied carbon and energy within the cradle-to-gate system boundary. However, this focus may overlook the potential environmental impacts of transportation (A4) and on-site construction activities (A5). For regions with limited forest resources, transportation (A4) plays a significant role in the environmental footprint of mass timber. Importing timber from other countries/regions often results in higher environmental impacts due to long transport distances, especially in where forest source is limited (Duan et al., 2022). On the other hand, on-site construction (A5) of mass timber buildings may offer benefits such as faster construction times, lower costs, and reduced emissions due to the high degree of prefabrication and installation efficiency, especially compared to concrete buildings (Jayalath et al., 2020). This aspect highlights the potential for overestimating or underestimating the environmental benefits of mass timber construction when these factors are not considered in embodied impacts.

An additional concern is the treatment of customized mass timber components in environmental impact assessments. Whether these impacts are included in building-level calculations is not always clear. This stage, defined as prefabrication in Takano et al. (2015), lies between the product stage and on site construction stage and should be explicitly stated for clarity. Incorporating considerations like localized energy mixes, transportation distances, and construction methods and efficiencies can improve the accuracy and reliability of mass timber LCA.

Finally, specific material properties and durability challenges associated with mass timber should also be addressed. The lower thermal mass of timber elements compared to concrete can result in slightly higher energy demands due to reduced thermal inertia and heat storage capacity (Dodoo et al., 2012). Furthermore, these materials may require enhanced maintenance against termite infestation, and degradation in high-moisture environments due to fungal decay (Ayanleye et al., 2022). In fact, it is reported that the Gaia building in NTU Singapore suffers mold issues due to the high humidity and temperature fluctuations (Built Offsite, 2024). In Australia, National Construction Code includes requirements of condensation management to minimize mould growth in buildings (Australian Building Codes Board, 2023), these regional climate considerations and associated maintenance requirements are not yet integrated in mass timber LCA. These contextual factors are typically not captured in standard database values but become significant in whole-building LCA. The gap between idealized database values and real-world performance metrics undermines confidence in current sustainability assessments. Without more reliable, context-specific data that captures these nuances, decision-makers risk basing material choices on incomplete or misleading information.

5. Recommendations and future directions for mass timber LCA development

The analysis of current challenges on LCA of bio-based materials reveals several important areas to enhance assessment accuracy, reliability and applicability for future directions of mass timber at material and building levels.

Establishing standardized assessment methods emerges as a fundamental priority, particularly concerning biogenic carbon accounting, system boundary definitions, and recycling allocation methodologies, which is crucial for consistent assessment of mass timber construction. While the importance of standardized assessment methods has been widely recognized in previous LCA studies and guidelines, this challenge still remains unresolved for bio-based materials. This is particularly problematic due to the dual role of mass timber as both a carbon sink and a potential source of emissions during its EoL. The current heterogeneity in carbon accounting approaches significantly impacts result reliability and comparability across studies. This necessitates methodological frameworks that systematically address both carbon uptake and release processes while incorporating temporal dynamics. Such standardization must extend to system boundary definitions and recycling calculations. In particular, clear protocols for allocating recycled content between the product stage and Module D credits are essential to prevent double-counting of environmental benefits. Moreover, current methodological frameworks inadequately address the complexity of EoL of mass timber, including options for component reuse, material recycling, energy recovery, and landfill disposal. The environmental impacts of these scenarios vary significantly from extended carbon storage through reuse to immediate carbon release through incineration. This variation creates the associated misalignment between environmental impact regarding the cascading uses versus immediate energy recovery which should be addressed in future direction for clear calculation.

Enhancing assessment accuracy for mass timber construction fundamentally relies on improving both material quantity precision and the reliability of environmental coefficients. First, the inherent complexity of mass timber constructions, characterized by intricate connections, specialized joining details, and auxiliary materials demands particular attention to modeling accuracy, which are often overlooked in current mass timber LCA. For instance, the utilization of oversimplified BIM models can lead to significant underestimation of environmental impacts. Simultaneously, environmental impact coefficients for mass timber require careful consideration of regional variations. Regional variations in manufacturing processes includes specific adhesive formulations, treatment methods, and localized production practices. Furthermore, future assessment must incorporate context-specific considerations that significantly influence the overall environmental profile: transportation impacts, prefabrication processes, installation efficiencies, and long-term maintenance requirements. Such details are typically unavailable for datasets from standard databases, however, they become increasingly crucial for developing reliable environmental assessments at the building scale. Future development must establish protocols that adequately capture these complexities, including quantitative precision, environmental coefficient reliability and overall life cycle stage details.

The development of LCA database and datasets selections highlight the need for enhanced transparency, open access, up-to-date and reliable data to address the unique challenges of mass timber assessment. Comprehensive datasets must document both fundamental parameters (e.g., density values) and detailed manufacturing specifications. Such datasets should maintain regular update cycles to reflect rapid advancements in processing technologies, treatment methods, and biogenic carbon accounting methods. Moreover, the LCA methodology shapes how databases represent and calculate environmental impacts of mass timber, greatly influencing the results. Process-based LCA, while offering detailed analysis of direct impacts, may underestimate total emissions due to system boundary truncation, particularly regarding biogenic carbon flows of mass timber. Input-output methods provide broader coverage but face significant temporal and spatial challenges. The economic data typically lags several years behind current practices, creating particular concerns for mass timber construction where industry practices are rapidly evolving. The spatial complexity increases further due to globalized mass timber supply chains, which challenges the selection between domestic, multi-regional, or global input-output tables. Hybrid approaches attempt to balance these trade-offs but introduce additional uncertainty and potential inconsistencies. The selection of appropriate LCA methodology for mass timber requires careful consideration of study objectives, data availability, supply chain complexity, and uncertainty tolerance.

Integration with industry practices emphasized the comprehensive material passport systems and broader forest management impacts. A significant challenge in current practice lies in matching actual materials used in construction with appropriate environmental data. Detailed specifications such as recycled content in steel, supplementary cementitious material proportions in concrete, or specific adhesive types in mass timber often remain undocumented or inaccessible for assessment. To address this challenge, future development should prioritize the implementation of integrated material passport systems within building models like BIM. This integration should encompass several key elements: (1) detailed material specifications that enable accurate matching between actual construction materials and environmental coefficients, (2) clear temporal and geographical metadata to ensure appropriate application of environmental data, and (3) advanced building models with construction assemblies (material layers, structure, etc.), connection systems and detailed disassembly protocols for future EoL reference. Such documentation systems support design optimization strategies that prioritize mass timber circularity through cascading principles. Furthermore, future research directions must develop frameworks to incorporate upstream carbon impacts from mass timber. This necessitates the integration of forest management practices and their temporal dynamics into mass timber LCA methodologies. Such research must evaluate the trade-offs between immediate carbon benefits achieved through material substitution versus the long-term forest carbon sequestration potential.

Future research priorities for mass timber construction highlight advancement in several critical domains. First, temporal carbon modeling frameworks must evolve to better capture the complex dynamics of EoL scenarios and their associated environmental implications. Current assessment methodologies inadequately address the fundamental misalignment between environmental benefits and circular economy principles, particularly concerning timber reuse strategies and biomass recovery pathways. Second, the research scope must expand beyond conventional carbon metrics to develop more comprehensive assessment frameworks. These frameworks should integrate multiple environmental dimensions: resource efficiency optimization, land use impact assessment, energy consumption patterns (both renewable and non-renewable), and broader environmental indicators. Moreover, there is a need to better understand the material circularity potential through design strategies and degradation patterns of mass timber in various climatic conditions. This requires investigation of the relationship between initial design of mass timber construction and long-term material circularity options. Particular attention must be paid to material aging mechanisms and their implications for structure and reuse potential.

6. Conclusions and outlook

This study provides a detailed analysis of how different LCA databases handle embodied GHG emissions in mass timber construction, revealing substantial methodological variations that affect environmental impact calculations. Through examination of five widely used databases, the research identifies and quantifies key factors driving assessment inconsistencies: biogenic carbon accounting methods, system boundary definitions, allocation approaches, and EoL scenarios.

The significant variations at both the material and building levels highlight methodological challenges and implications for LCA assessment practices. Mass timber exhibits the most extreme variability among construction materials, with emissions ranging from −686.80 to 1718.00 $kgCO_2eq/m^3$, primarily driven by divergent approaches to biogenic carbon accounting and varying methodological frameworks. At the building level (A1-A3), mass timber buildings show wider variation (−170.31 to 434.62 $kgCO_2eq/m^2$) compared to RC buildings (69.79 to 485.95 $kgCO_2eq/m^2$), underscoring how dataset selection and accounting methods can significantly impact the carbon footprint of mass timber construction.

The research identifies several critical challenges in current LCA practices for bio-based materials/construction. The hybrid method employed in databases like EPiC yields notably higher emissions values, particularly for mass timber products, highlighting the influence of methodological choices on results. Inconsistent treatment of biogenic carbon flow across databases, varying system boundary definitions, and inadequate consideration of EoL processes create substantial obstacles to accurate assessment. When considering biogenic carbon, a consistent approach is essential. If biogenic carbon uptake is accounted for, the release stage must also be included to avoid inappropriate results. Temporal considerations and data timeliness significantly affect LCA assessment reliability, especially for rapidly evolving materials like mass timber. Moreover, while carbon emissions often dominate sustainability discussions, comprehensive assessments must consider multiple indicators, including energy, economic factors, and material efficiency.

The significance of EoL modeling for timber products lies in two reasons: it reflects material efficiency and up/down-cycling potential, and it captures the complete biogenic carbon flow if modeled. Simply modeling A1-A3 stages with biogenic carbon sequestration will greatly overestimate the benefits of mass timber construction. In fact, using timber to replace high carbon intensive materials versus the reduced timber harvesting that allows forests to grow and absorb carbon has raised great concerns. Neglecting biogenic carbon may also underestimate the carbon sequestration of these materials and carbon sink potential at a larger city scale. Using mass timber data from EPDs requires careful evaluation, particularly regarding Module D calculations, where significant variations exist in methodologies and underlying assumptions. These variations become especially problematic when combining EPD data with database data for building-level assessment, as inconsistencies in system boundaries and modeling approaches can lead to unreliable results.

Regional variations in mass timber manufacturing processes, energy sources, and transportation distances emerge as essential factors influencing embodied impacts. The study emphasizes that using context-appropriate and updated databases depends on the research scope. Significant disparities warrant careful attention even within the same region, as demonstrated by the differences between EPiC and AusLCI in the Australian context. The data transparency and consideration of prefabrication processes in mass timber also raise concerns about the data quality. Future development of datasets for the bio-based material should prioritize comprehensive material data, including specific properties such as density, detailed information about prefabrication processes and auxiliary materials, and expanded coverage of EoL scenarios with standardized calculation methods. The study highlights maintenance implications and EoL calculation reliability, particularly in regions with challenging climatic conditions where mass timber performance may vary greatly.

To enhance the reliability of mass timber embodied impacts, this research recommends establishing standardized approaches for biogenic carbon accounting and consistent system boundaries that consider both the uptake and release of the biogenic carbon. To capture the environmental impacts, both the quantity of building materials and the quality of building materials value need consideration. Oversimplified BIM may underestimate the building material quantities. Untransparent documentation of material specifications and manufacturing processes in databases leads to inaccuracies in LCA. Another essential challenge is the difficulty researchers face in selecting material datasets that accurately reflect the real local context of a building project. This issue might be addressed by implementing detailed material passports for building materials, integrated with models like BIM and environmental assessment tools. Such an approach would not only improve LCA accuracy but also enable more effective modeling of EoL recycling scenarios, ultimately enhancing mass timber circularity.

In addition, focusing on modeling recycling scenarios and their implication emissions is not enough for mass timber construction, as the early design strategies and principles would have a greater impact on the waste amount and quality of the timber. These early design decisions, once made, are difficult to modify but continue to influence subsequent phases. Several key strategies emerge from this perspective: First, design for disassembly becomes paramount, enabling timber elements to be recovered and reused across multiple building life cycles, thereby extending the duration of carbon storage while reducing harvest pressure on forests. Second, implementing cascading principles would maximize the utility derived from each unit of harvested timber before its EoL. Third, standardization of connection systems and element dimensions could facilitate future reuse, creating a more robust circular economy for mass timber components. This highlights future collaboration from different stakeholders to integrate design process, manufacturing, and construction considerations, as current practices often handle these aspects separately.

However, this research also suffers from several limitations. First, the analysis encompasses five databases representing specific geographical contexts, which may limit the generalizability of findings to regions not represented in the study. Second, the research focus was limited to carbon metrics. Future studies should expand to multiple environmental impact indicators, including energy, water, ODP, and AP, etc. The analysis encompasses a single 10-storey office building case study, which may limit the generalizability of findings across different building types, heights, and structural designs. Future research should extend this analysis to a range of building archetypes to validate the robustness of our findings across diverse construction contexts. Furthermore, this research employs simplified BIM modeling approaches that may not fully capture the complexity of structural design and "as-built" material quantities in real-world construction, particularly for mass timber elements with intricate connections and auxiliary components. Recent advances in data-driven methodologies may provide more realistic embodied carbon estimates than the idealized models typically used in current practice. Finally, the misalignment between environmental benefits and circular economy principles in mass timber reuse and energy recovery pathways remains unaddressed.

The study concludes that while mass timber presents promising opportunities for reducing construction-related emissions, accurate assessment of its environmental benefits requires careful consideration of datasets/database selection, methodological consistency, and regional context. The research identifies several persistent challenges in LCA, including data variability and reliability, result incomparability, and the need for consistent and transparent databases and methodologies for all building materials. Future development of LCA databases should prioritize transparency, consistency, and comprehensiveness of building materials and localization. Future work must balance material substitution benefits with forest resource preservation. Success in these developments demands sustained collaboration among research institutions, industry practitioners, building designers, and policy makers. Those understandings will be essential for supporting the transition toward more sustainable building practices and realizing the full potential of mass timber in mitigating climate change.

CRediT authorship contribution statement

Zhuocheng Duan: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Hossein Omrany:** Writing – review & editing, Supervision. **Jian Zuo:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Table A1

Area/ Countries	Focus	Indicators	Date sources (mass timber)	Ref.
Korea	Concrete/Exterior CLT walls/CLT floor/CLT roofs/all CLT	GHG emissions	References	(Kang et al., 2024)
U.S.	Fireproofing CLT/ Charring CLT/ Concrete	GWP, ODP, EP, AP, Smog	CORRIM Report and references	(Pierobon et al., 2019)
Malaysia	GLT/CLT (Hardwood and softwood)	GWP, ODP, Terrestrial Ecotoxicity, HTP, FDP, Land use potential, Embodied energy	Modified based on Laminated timber element from SimaPro database	(Balasbaneh and Sher, 2021)
Sweden	(Conventional and low energy) Beam-and-column/ CLT/Modular buildings	Life cycle GHG emissions	Ecoinvent, reference	(Dodoo et al., 2014a)
Sweden	Beam-and-column/CLT/Modular buildings	Life cycle primary energy	Ecoinvent, reference	(Dodoo et al., 2014b).
Sweden	Concrete/CLT/Modular	Life cycle GHG emissions, Life cycle primary energy	Ecoinvent	(Piccardo and Gustavsson, 2021)
China	CLT/RC	GWP	Athena	(Liu et al., 2016)
Australia	Post tensioned concrete /Mass Timber building	Embodied carbon	References, ICE, EPiC	(Robati and Oldfield, 2022)
Australia	CLT/RC	Life cycle GHG emissions	EPiC	(Jayalath et al., 2020)

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Area/ Countries	Focus	Indicators	Date sources (mass timber)	Ref.
Finland	light-weight timber/ massive timber/ precast concrete	GHG emission	GaBi, Ecoinvent, IBO, CFP, Synergia	(Takano et al., 2014b)
Sweden	CLT/Increased bio/Concrete	GWP using Dynamic LCA	Modified based on Ecoinvent 2.0	(Peñaloza et al., 2016)
Italy	CLT/RC/Steel	Thermal comfort, life cycle energy/GHG emissions, life cycle cost	ICE, Ecoinvent 3, EPD	(Invidiata et al., 2018)
Germany	Timber (timber frame, CLT, prefabricated timber)/ Mineral (RC, porous concrete, brick, sand-lime brick) buildings	GWP (biogenic and fossil)	oekobau.dat	(Hafner and Schäfer, 2017)
U.S.	Mass timber/precast concrete/steel/post-tensioned concrete	Embodied carbon/energy	ICE	(Zeitz et al., 2019)
Finland	light weight timber/ CLT/RC/Aircrete/Brick/ Steel	Resource use, embodied GHG/energy, carbon storage, cost, energy content	Ecoinvent	(Takano et al., 2014a)
UK	CLT/concrete/steel/load-bearing masonry	Embodied carbon	Reference	(Moncaster et al., 2018)
Norway	CLT/RC	GWP with biogenic carbon	Ecoinvent	(Andersen et al., 2022)
UK	Steel/RC/engineered timber	Embodied carbon	SimaPro, EPD	(Hart et al., 2021)

Data availability

Data will be made available on request.

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